Xiaowen Qiu

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Education

- **Fudan University** — Shanghai, China
  Bachelor of Science - Computer Science; GPA: 3.3/4.0 — *September 2019 - June 2023*
  Courses: *Calculus, Linear Algebra, Probability, Operating System, Data Structure, Introduction to Algorithms, Artificial Intelligence, Pattern Recognition and Machine Learning, Computer Network, Introduction to Database*

- **University of Massachusetts, Amherst** — Amherst, MA
  Master of Science - Computer Science; GPA: 3.7/4.0 — *September 2023 - May 2025*
  Courses: *Machine Learning, Neural Networks: Modern Intro, Logic in Computer Science, System Defense and Test, Mobile and Ubiquit. Computing, Software Engineering, Game Programming, Computer Vision, Robotics*

- **University of Massachusetts, Amherst** — Amherst, MA
  Ph.D. - Computer Science — *Sep 2026 – Present*

Publications

- **Gen2Dex: Learning Dexterous Manipulation from Simulation-Initialized Generated Video Demonstrations** — arXiv 2026
  Xiaowen Qiu*, Yian Wang, Haoxiang You, Chuang Gan*
  Under Review at ICRA 2027
- **IM-ENGINE: Image Editing for Embodied Data Generation** — CoRL 2026
  Yian Wang, Junyi Cao*,* ***Xiaowen Qiu****, Chuang Gan*
- **PhyScensis: Physics-Augmented LLM Agents for Complex Physical Scene Arrangement** — ICLR 2026
  Yian Wang, Han Yang*, Minghao Guo,* ***Xiaowen Qiu****, Tsun-Hsuan Wang, Wojciech Matusik, Joshua B. Tenenbaum, Chuang Gan*
- **Robotsmith: Generative robotic tool design for acquisition of complex manipulation skills** — NeurIPS 2025
  Chunru Lin, Haotian Yuan*, Yian Wang*,* ***Xiaowen Qiu****, Tsun-Hsuan Wang, Minghao Guo, Bohan Wang, Yashraj Narang, Dieter Fox, Chuang Gan*
- **Articulate AnyMesh: Open-Vocabulary 3D Articulated Objects Modeling** — CoRL 2025
  Xiaowen Qiu*, Jincheng Yang*, Yian Wang, Zhehuan Chen, Yufei Wang, Tsun-Hsuan Wang, Zhou Xian, Chuang Gan*
- **Architect: Generating Vivid and Interactive 3D Scenes with Hierarchical 2D Inpainting** — NeurIPS 2024
  Yian Wang,* ***Xiaowen Qiu****, Jiageng Liu*, Zhehuan Chen, Jiting Cai, Yufei Wang, Tsun-Hsuan Wang, Zhou Xian, Chuang Gan*
- **3D-VLA: A 3D Vision-Language-Action Generative World Model** — ICML 2024
  Haoyu Zhen, ***Xiaowen Qiu****, Peihao Chen, Jincheng Yang, Xin Yan, Yilun Du, Yining Hong, Chuang Gan*

Experience

- **Fudan Nebula AI LAB**
  Undergraduate Researcher, supervised by Prof. Weifeng Ge — *Mar 2021 - Jun 2023*
  Developed a cycle-consistent model for photorealistic style transfer. Built a voxel diffusion model for semantic scene completion.

- **Microsoft Research Lab - Asia (MSRA)**
  Research Intern — *Apr 2022 - Mar 2023*
  Implemented DINO-based models for table detection and scene text detection, improving detection accuracy.

- **UMass Embodied AGI Lab**
  Graduate Researcher, supervised by Prof. Chuang Gan — *Sep 2023 - May 2025*
  Embodied AI and Robotics (3D-VLA; Architect for Interactive 3D Scene Generation; Articulate AnyMesh for Open-Vocabulary 3D Articulated Object Modeling; Robotsmith; PhyScensis)

- **Genesis Intelligence Inc.**
  Research Scientist Intern — *June 2025 - Aug 2025*
  Built a synthetic robot-data augmentation pipeline using MimicGen and other augmentation methods. Ran experiments on co-training robot policies with simulated and real-world data.

- **Genesis Intelligence Inc.**
  Research Scientist — *Aug 2025 - Aug 2026*
  Built infrastructure for simulation-based policy evaluation. Contributed to the GENE-26.5 release by delivering the wire-harnessing demo task and robot foundation-model evaluation infrastructure and results. Led the development of a hardware stack for glove-based data collection.

- **UMass Embodied AGI Lab**
  Graduate Researcher, supervised by Prof. Chuang Gan *Sep 2026 - Present*
  Working on Agentic Robotics

Projects

- **Genesis-World**: A physics and robotics simulation engine. Project page: https://genesis-embodied-ai.github.io/.